Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2023-P1-Model)** **Date: 08-02-2026**

Time: 09.00Am to 12.00Pm **GTA-15** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**ABCD**	2	**AB**	3	**AC**	4	**C**	5	**B**	6	**B**
7	**B**	8	**1**	9	**1**	10	**3**	11	**36**	12	**4**
13	**5**	14	**B**	15	**C**	16	**D**	17	**B**		

PHYSICS

18	**BCD**	19	**A**	20	**A**	21	**C**	22	**C**	23	**C**
24	**B**	25	**7**	26	**5**	27	**5**	28	**1**	29	**1**
30	**8**	31	**C**	32	**A**	33	**A**	34	**C**		

CHEMISTRY

35	**ABCD**	36	**AC**	37	**BCD**	38	**A**	39	**D**	40	**C**
41	**C**	42	**7**	43	**4**	44	**33**	45	**5**	46	**8**
47	**12**	48	**B**	49	**D**	50	**A**	51	**A**		

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SOLUTIONS
MATHEMATICS

1. Let $m_1 = \text{slope of BC} = \dfrac{2}{t_2 + t_3}$

And $m_2 = \text{slope of AD} = \dfrac{0 - 2t_1}{1 - t_1^2} = \dfrac{2t}{t_1^2 - 1}$

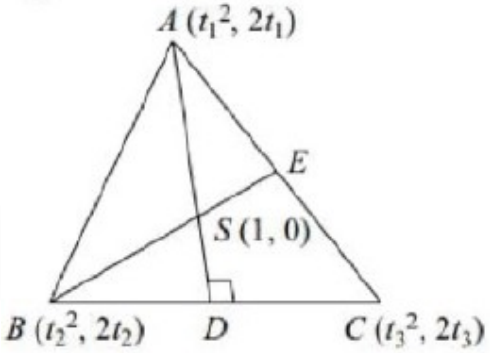
As $AD \perp BC$, $m_1 m_2 = -1 \quad \Rightarrow t_2 + t_3 - 4t_1 = t_1^2 t_2 + t_1^2 t_3$ (i)

Similarly, $t_3 + t_1 - 4t_2 = t_2^2 t_3 + t_1^2 t_1$ (ii)

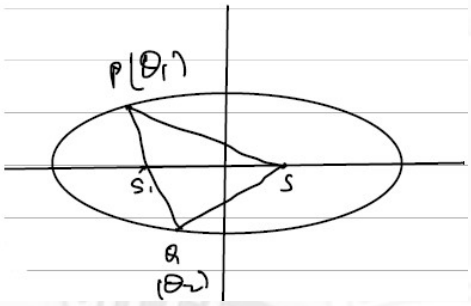
From (i) $\Rightarrow t_1(t_1 t_2 + t_1 t_3) = t_2 + t_3 - 4t_1 \Rightarrow t_1[-5 - t_2 t_3] = t_2 + t_3 - 4t_1$

$\Rightarrow \dfrac{1}{t_2 t_3} + \dfrac{1}{t_3 t_1} + \dfrac{1}{t_1 t_2} = -1$

Also, $(1 + t_1)(1 + t_2)(1 + t_3) = 1 + t_1 + t_3 + (t_1 t_2 + t_2 t_3 + t_3 t_1) + t_1 t_2 t_3 \quad = 1 + 0 - 5 = -4$



2. Sol – 'P' na 'Q' should be the endpoints of one of the focal chord to make the perimeter of ΔPQS constant.



$\therefore PS + SQ + PQ = 4a \quad \Rightarrow 2r = 4a$; r is semi perimeter of $\Delta PQS \quad \Rightarrow r = 2a$

Now, $A = \sqrt{r(r - PS)(r - PQ)(r - QS)}$

$A = \sqrt{-2a^4(1 + e\cos\theta_1)(1 + e\cos\theta_2)(\cos\theta_1 + \cos\theta_2)}$ $\quad \therefore PQ$ is a focal chord.

$\therefore \cos\left(\dfrac{\theta_1 - \theta_2}{2}\right) = -e\cos\left(\dfrac{\theta_1 + \theta_2}{2}\right)$

Also, $\dfrac{2PS_1 \cdot QS_1}{PS_1 + QS_1} = \dfrac{b^2}{a} \quad \Rightarrow PS_1 . QS_1 = b^2(2 + e(\cos\theta_1 + \cos\theta_2))$

$\therefore a^2(1 + e\cos\theta_1)(1 + e\cos\theta_2)$

$\Rightarrow (1 + e\cos\theta_1)(1 + e\cos\theta_2) = \dfrac{b^2}{a^2}(2 + e(\cos\theta_1 + \cos\theta_2))$

$\therefore A = \sqrt{2}eab\sqrt{-(2 + e(\cos\theta_1 + \cos\theta_2))(\cos\theta_1 + \cos\theta_2)}$

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3. $f(x) = ax^2 + bx + c - b$

Given , $\frac{-b}{2a} = 0 \Rightarrow b = 0$

Now, let maximum value of g(x) occur at x = p

$\Rightarrow g'(p) = 0 \quad \Rightarrow \left[(f'(p))^2 + f''(p)\right] e^{f(p)} = 0$

$\Rightarrow \left[(f'(p))^2 + f''(p)\right] = 0 \qquad \ldots.(i)$

Also , g(x) = $4\sqrt{e}$ has rational roots (given)

$\therefore |f'(p)|\, e^{f(p)} = 4\sqrt{e}$ has rational root (p must be retional)

On comparing, $|f'(p)| = 4$ and $f(p) = \frac{1}{2} \quad \ldots(ii)$

From (i) & (ii),

$16 + 2a = 0 \Rightarrow a = -8$

Hence, f(x) = $-8x^2 + c$

Now, f(p)=$\frac{1}{2} \Rightarrow -8p^2 + c = \frac{1}{2}$

And $f'(p) = -16p = \pm 4 \qquad \Rightarrow p = \pm\frac{1}{4}$

Hence, c = 1 $\therefore f(x) = -8x^2 + 1$

g(x) = $|16x|\, e^{1-8x^2}$ Now verify options

4. $a_1a_2a_3a_4$

$b_1b_2b_3b_4$

$c_1c_2c_3c_4$

$d_1d_2d_3d_4$

Now for unit's place:-

$a_4 + b_4 + c_4 + d_4 < 10$

$a_4 \geq 0, b_4 \geq 0, c_4 \geq 0, d_4 \geq 0$

$a_4 + b_4 + c_4 + d_4 \leq 9$

For ten's and hundred's place:-

No of solution - $^{13}C_4$

For thousand's place :-

$a_1 + b_1 + c_1 + d_1 \leq 9$

$\therefore a_1 + b_1 + c_1 + d_1 + u = 9, \qquad u \geq 0$

$\therefore$ Total no. of required numbers : - $\left(^{13}C_4\right)^3\, {}^9C_4$

$\therefore$ Total no. of required numbers : - $\left(^{13}C_4\right)^3\, {}^9C_4$

5. Differiatiny wrt 'x'

$$e^n + e^{-x}f(x) - e^{-x}\int_0^x f(t)dt = \frac{\int_0^x e^x f(x)dx}{e^x}$$

$$\Rightarrow e^{2x} + f(x) - \int_0^x f(t)dt = \int_0^x e^x f(x)dx$$

Again differentiate w.r.t 'x'

$2e^{2x} + f'(x) - f(x) = e^x f(x)$

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$\Rightarrow \frac{dy}{dx} - \left(1+e^x\right)y = -2e^{2x} \qquad \therefore y = 2e^x + \frac{1}{e}e^x.e^{e^x}$

6. $f(t) = \lim\limits_{z\to\infty} z^k \int\limits_0^{1/z} t^{t+k-1}dt$

Put $Z = \frac{1}{x}$ $= \lim\limits_{x\to 0} \frac{\int\limits_0^x t^{t+k-1}}{x^k}$ $= \lim\limits_{x\to 0} \frac{x^x \cdot x^{k-1}}{kx^{k-1}}$ $= \frac{1}{k}\lim\limits_{x\to 0} x^x$

$= \frac{1}{k}\lim\limits_{x\to 0} e^{x\ln x}$ $= \frac{1}{k}\lim\limits_{x\to 0} e^{\frac{\ln x}{x}}$ $= \frac{1}{k}$

$f(4) = \frac{1}{4}$ $xy = 4,\ x^2 + y^2 = 8$

$x^2 + \frac{16}{x^2} = 8$ $x^4 - 8x^2 + 16 = 0$ $x^2 = 4 \Rightarrow x = \pm 2$ $y = \pm$

"Two comman Points"

7. $1 - x^3 = t \qquad \Rightarrow -3x^2dx = dt$

$3I_1 = \int\limits_0^1 \left(1-t^{\sqrt{2}}\right)^{\sqrt{3}} dt$

$3I_2 = \int\limits_0^1 \left(1-t^{\sqrt{2}}\right)^{\sqrt{3}+1} dt$

Now $3I_2 = \int\limits_0^1 (1-t^{\sqrt{2}})^{\sqrt{3}+1} 1\,dt$

Apply by parts

8. $e^{-x}f(x) = f'(x) + f''(x) + f'''(x) + \ldots\ldots\ldots\infty$

Differentiable both sides w.r.t 'x'

$\Rightarrow e^{-x}f'(x) - e^{-x}f(x) = f''(x) + f'''(x) + \ldots\ldots\ldots\infty$

$\Rightarrow e^{-x}f'(x) - e^{-x}f(x) = e^{-x}f(x) - f'(x)$

$\Rightarrow f'(x)\left(1+e^{-x}\right) = 2e^{-x}f(x)$

$\Rightarrow \frac{f'(x)}{f(x)} = \frac{2e^{-x}}{1+e^{-x}} \qquad \Rightarrow \int \frac{df(x)}{f(x)} = \int \frac{2e^{-x}}{1+e^{-x}}dx$

$\Rightarrow \ln|f(x)| = \ln\frac{1}{\left(1+e^{-x}\right)^2} \qquad \Rightarrow f(x) = \frac{1}{\left(1+e^{-x}\right)^2} \qquad \therefore \lim\limits_{x\to\infty} f(x) = 1$

9. $k_1 = \lim\limits_{t\to\infty} t\int\limits_0^1 x^t g(x)\,dx\,;$ $k_2 = \lim\limits_{t\to\infty} t\int\limits_0^1 x^t h(x)\,dx$

$e^{4x}k_1 \le f''(x) - 4f'(x) + 3f(x) \le 2e^{5x}k_2$

$\Rightarrow e^x k_1 \le e^{-3x}\left(f''(x) - f'(x) - 3\left(f'(x) - f(x)\right)\right) \le 2e^{2x}k_2$

10. let $g(x) = \int\limits_o^x f(t^2)dt$ then $g(-x) = \int\limits_o^{-x} f(t^2)dt$

let t = -u

dt = -du $\therefore g(-x) = -g(x)$

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Substitute $x \to -x$

$$\frac{f(x^2)}{1+e^{-x}}+\frac{f'(x^2)}{1-e^{-x}}+x^2=\int_{g(x)}^{-g(x)} f(t)dt$$

$$\Rightarrow \frac{e^x f(x^2)}{1+e^x}+\frac{e^x f'(x^2)}{e^x-1}+x^2=\int_{g(x)}^{-g(x)} f(t)dt \quad ---(1)$$

$$\frac{f(x^2)}{1+e^x}-\frac{f'(x^2)}{e^x-1}+x^2=\int_{g(x)}^{g(x)} f(t)dt \quad ----(2)$$

$1 + 2$

$$f(x^2)+f'(x^2)+2x^2=0$$

Replace $x^2 = t$

$$F(t)+f'(t)+2t=0$$

$$\Rightarrow e^t\left(f(t)+f'(t)\right)=-2te^t \Rightarrow \frac{d}{dt}(e^t f(t))=-2te^t$$

$$\Rightarrow \int d(e^t f(t))=-2te^t dt \Rightarrow f(t)=2+e^{-t}-2t$$

11. $$I_n=\int_0^1 x^n \cot^{-1}x\,dx,\ I_n=\cot^{-1}x.\frac{x^{n+1}}{n+1}-\int\frac{-1}{1+x^2}.\frac{x^{n+1}}{x^2+1}dx$$

$$(n+1)I_n=\left[\cot^{-1}x.x^{n+1}\right]_0^1+\int\frac{x^{n+1}}{x^2+1}dx \quad (n-1)I_{n-2}=\left[\cot^{-1}x.x^{n-1}\right]_0^1+\int\frac{x^{n-1}}{x^2+1}dx$$

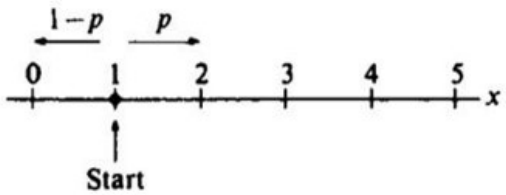
$$(n+1)I_n+(n-1)I_{n-2}=\frac{\pi}{2}+\int_0^1 x^{n-1}dx \qquad =\frac{\pi}{2}+\frac{1}{n}$$

$$13I_2+11I_{10}=\frac{\pi}{2}+\frac{1}{12} \quad (n+1)J_n+(n-1)J_{n-2}=\frac{\pi}{2}-\frac{1}{n} \quad 10J_9+8J_7=\frac{\pi}{2}-\frac{1}{9}$$

$$13I_{12}+11I_{10}+10J_9+8J_7=\pi+\frac{1}{12}-\frac{1}{9} \qquad =\pi\frac{-3}{108} \qquad =\pi-\frac{1}{39}$$

$m = 1 \quad n = 36 \qquad \left[\frac{n}{m}\right]=36$

12. If the particle ever arrives at x = 0, it is absorbed there (takes no further steps). We wish to know the probability, P_1, that the particle is absorbed at x = 0, given that it starts at x = 1.



Consider the situation after the first step: either the particle moved left to x = 0 and was absorbed (this event has probability 1 – p) or it moved right to x = 2 (this event has probability p). Let P_2 be the particle starts from position x = 2. Then we can write

(1) $P_1 = 1 - p + pP_2$

Because 1 – p is the probability of absorption at the first step and pP_2 is the probability of being absorbed later. The probability of a path from x = 2 to x = 1 is just P_1, because the structure here is identical with that of the original set-up for the particle except that the

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origin has been translated one step to the right. The probability of a path from x = 1to x = 0 is also P_1, because this is exactly the original problem. The probability P_2 therefore is P_1^2, because the events A = (particle takes path from x = 2 to x = 1) and B = (particle takes path from x = 1 to x = 0) are independent, and P(A) = P(B) = P_1

We can rewrite eq. (1) as $P_1 = 1 - p + pP_1^2$

13. $y = 2 + (1-x) - \sqrt{x^2 - 2x + 1}, x < 1 \quad = 2 + (1-x) - (1-x)\ y = 2$

A, $y = 2\ x^2 + 4 = 13\ x^2 = 9\ x = 3, -3, x < 1, \quad x_1 = -3$

A (-3, 2, 0) $x_2^2 + y_2^2 < 13\ x^2 < 9 \quad -3 < x_2 < 3, \quad x_2 < 1$

Max $(x_2) = 0$

Equation of line $\frac{x+3}{3} = \frac{y-2}{4} = \frac{z-3}{5}$

$3(3\lambda) + 4(4\lambda) + 5(5\lambda + 3) = 0\ 50\lambda = -15\ \lambda = \frac{-3}{10}$

14. (I) $S = \sum_{k=1}^{n} (-1)^{k-1}\ {}^nC_k \left(1 + \frac{1}{2} + \frac{1}{3} + \ldots\ldots + \frac{1}{k}\right) = \sum_{k=1}^{n} \left((-1)^{k-1}\ {}^nC_k \sum_{k=1}^{n} \int_0^1 x^{r-1} x\right)$

$= -\int_0^1 -\frac{(1-x)^n}{1-x} dx \quad = \int_0^1 (1-x)^{n-1} dx = \int_0^1 x^{n-1} dx = \frac{1}{n}$

(II) $S = \sum_{r=0}^{9} \frac{{}^9C_r (-1)^r}{r+8}\ x^7 (1-x)^9 = \sum_{r+8}^{9} {}^9C_r (-1)^r\ x^{r+7}$

$\int_0^1 x^7 (1-x)^9 dx = \sum_{r=0}^{9} \frac{{}^9C_r (-1)^r}{r+8} = \frac{7!9!}{17!} = \frac{7\times6\times5\times4\times3\times2}{17\times16\times15\times14\times13\times12\times11\times10} = \frac{1}{17\times16\times13\times11\times5}$

(III) $N = 43^{43^{43}} \quad 43^{43} = (40+3)^{43} = 4k + 3\ N = 43^{4k+3} = (40+3)^{4k+3} = 4\lambda + 81^k.27$

Remainder = 27

(IV) $S_n = \sum_{k=0}^{n} \frac{{}^{n+k}C_n}{2^k}$ = coefficient of x^n in

$\left((1+x)^n + \frac{(1+x)^{n+1}}{2} + \frac{(1+x)^{n+2}}{2^2} + \ldots\ldots + \frac{(1+x)^{n+n}}{2^n}\right)$

$S_n = 2^n \Rightarrow S_{12} = 4096 \quad \frac{S_n}{40} \Rightarrow$ Remainder = 16

15. (I) $f(x_1) > f(x_2) \Rightarrow N = {}^{8+4-1}C_{4-1} = {}^{11}C_3 = 165 \Rightarrow 12$

(II) $f(x_1) \le f(x_2) \Rightarrow N = {}^{8-1}C_{4-1} = {}^7C_3 = 35 \Rightarrow 8$

(III) $N = {}^8C_4 \times 4! - {}^4C_1 . {}^7C_3 3! + {}^4C_2 . {}^6C_2 \times 2! - {}^4C_3 . {}^5C_1 + 1 = 1001 \Rightarrow 2$

(IV) $N = {}^4C_2 \times [5\times5 - 4] = 126 \Rightarrow 9$

16. (I) Let $g(x) = f(x) e^{\cos x} \quad \therefore g(x)$ is decreasing function

$\therefore g(x) \le g(0) = 0$ i.e. $f(x) e^{\cos x} \le 0$ but $f(x) \ge 0,\ e^{\cos x} > 0 \Rightarrow f(x) = 0 \quad \therefore f\left(\frac{3\pi}{2}\right) = 0$

(II) $(a-3)x^2 + 12x + 6 + a - \left(\sin^{-1}\sin 100 + \cos^{-1}\cos 100\right) < 0 \quad \Rightarrow a < 3.D < 0$

$144 - 4(a+6)(a-3) < 0 \Rightarrow a < -9$ or $a > 6$.

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(III) $P\left(\frac{B}{A\cup B^C}\right)=\frac{P(A\cap B)}{P(A\cap B^C)}=\frac{1}{5}$

(IV) $\cos^{-1}\left(4x^3-3x\right)-\frac{\pi}{2}\geq 0 \Rightarrow -1\leq 4x^3-3x\leq 0$ i.e. $x\in\left[-1,-\frac{\sqrt{3}}{2}\right]\cup\left[0,\frac{\sqrt{3}}{2}\right]$

also $\frac{(2x+3)!}{\sqrt{x+2}}$ is defined $\forall x>-2$ and $(2x+3)\in W$ $x=-1,0,\frac{1}{2}$

17. (I) Equation of circle is $x^2+\left(y-\frac{10}{3}\right)^2=1$

Consider a general point $\left(2t^2,2t^3\right)$ on the curve $x^3=2y^2$, its distance from centre is

$$\ell^2=4t^4+\left(2t^3-\frac{10}{3}\right)^2=4\left[t^4+\left(t^3-\frac{5}{3}\right)^2\right]$$

$\frac{d(\ell)^2}{dt}=8t^2(t-1)\left(3t^2+3t+5\right)\Rightarrow t=1$ is a point of local minima $\Rightarrow \ell_{min}=\frac{2}{3}\sqrt{13}$

(II) Let $t=-x-\frac{\pi}{6}, t\in\left[\frac{\pi}{6},\frac{\pi}{4}\right]$ $\Rightarrow \tan\left(x+\frac{2\pi}{3}\right)-\tan\left(x+\frac{\pi}{6}\right)+\cos\left(x+\frac{\pi}{6}\right)$

$\Rightarrow$ Greatest value occur at $t=\frac{\pi}{6}\Rightarrow\frac{11\sqrt{3}}{6}$

(III) $I_1=\lim_{x\to\infty}-x^2\left(\left(1+\frac{1}{x}+\frac{1}{x^3}\right)^{\frac{1}{3}}+\left(1-\frac{1}{x}+\frac{1}{x^3}\right)^{\frac{1}{3}}-2\right)$ $=\lim_{x\to\infty}-x^2\left((1+\alpha)^{\frac{1}{3}}+(1-\beta)^{\frac{1}{3}}-2\right)$

$\left(\text{Where, } \alpha=\frac{1}{x}+\frac{1}{x^3},\beta=\frac{1}{x}-\frac{1}{x^3}\right)$

$=\lim_{x\to\infty}-x^2\left(\frac{2}{3x^3}-\frac{1}{3}\left(\frac{2}{x^3}+\frac{2}{x^3}\right)+\ldots\ldots\ldots\right)=\frac{2}{9}$

(IV) Put n = 1; $f'(x)=f(x+1)-f(x), n=2;$

$f'(x)=\frac{f(x+2)-f(x)}{2}$

So, $f'(x)=\frac{f(x+2)-f(x+1)+f(x+1)-f(x)}{2}$,

$f'(x)=\frac{1}{2}, f'(x+1)\frac{1}{2}f'(x)$

$\Rightarrow (f(x+1))-f(x)=0\forall x\in R$

$\Rightarrow (f(x+1))-f(x)=c$ for a constant $c\in R$

$\Rightarrow f'(x)=c\ \therefore c=2$

$f(0)=d=3\ \therefore f(x)=2x+3$

$\frac{f(6)}{f(1)}=3$

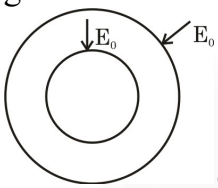
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PHYSICS

18. $J=\sigma E=\frac{1}{\rho}E \qquad J=(a+bh^2)E_0$

Hence option (A) is incorrect. Current density increases with increase in height. As field is directed downward electrons will flow upward (radially outward).

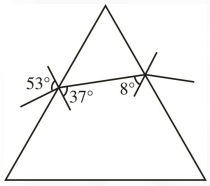
Now, to calculate charge density (ρ) we use gauss law (Assume a thin spherical shell as gaussian surface)



$$\phi=\phi_{IN}+\phi_{out}=\frac{q}{\varepsilon_0}$$

$$\Rightarrow -E_0\left(4\pi r^2\right)+E_0 4\pi\left(r-dr\right)^2=\frac{\rho\left(4\pi r^2 dr\right)}{\varepsilon_0}\Rightarrow \rho\approx -\frac{2\varepsilon_0 E_0}{r}=-\frac{2\varepsilon_0 E_0}{R+h}$$

19.



Angle of incidence is the brewster's angle $\therefore \tan i=\mu=\frac{4}{3}\Rightarrow i=53^o$

Snell's law at the first interface $\Rightarrow 1\times\sin 53^o=\frac{4}{3}\sin r_1\Rightarrow r_1=37^o$

Using relation $r_1+r_2=A\Rightarrow r_2=45^o-37^o=8^o$

Snell's law at the second interface $\frac{4}{3}\sin 8^o=1\times\sin e$

$$\Rightarrow\frac{4}{3}\left(8^0\times\frac{\pi}{180}\right)=e\times\frac{\pi}{180}\{\because\sin\theta\approx\theta\}\Rightarrow e=\frac{32}{3}\text{ degree}$$

Using $i+e=\delta+A \qquad \Rightarrow\delta=53+\frac{32}{3}-45=18.7^o$

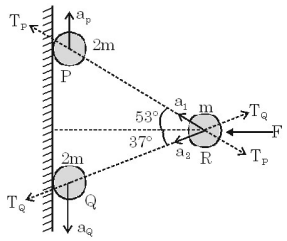
Reflected ray gets polarized. Transmitted ray remains as it is in terms of polarization. For no partial reflection light must be incident at brewster's angle and incident light must be zolarized in a particular direction.

20. From the given situation,

$$F-\alpha V^2=mv\frac{dv}{dx}\Rightarrow\frac{F-\alpha V^2}{F-\alpha V_0^2}=e^{-2\alpha x/m}\therefore F=a\left[\frac{v^2-v_0^2e^{-2\alpha x/m}}{1-e^{-2ax/m}}\right]$$

21.

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Let component of acceleration of disc R are a_1 and a_2

$$a_1 = a_p \cos 37^o = \frac{4a_p}{5} \qquad \ldots(1)$$

$$a_2 = a_Q \cos 53^o = \frac{3a_Q}{5} \qquad \ldots(2)$$

For P $T_p \cos 37^o = 2ma_p \quad \frac{4T_P}{5} = 2ma_P \qquad \ldots(3)$

For Q $T_Q \cos 53^o = 2ma_Q \quad \frac{3T_Q}{5} = 2ma_Q \qquad \ldots(4)$

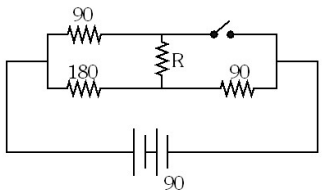
For R $F \cos 53^o - T_P = ma_1 \quad \frac{3F}{5} - T_P = ma_1 \qquad \ldots(5)$

and $F \cos 37^o - T_Q = ma_2 \quad \frac{4F}{5} - T_Q = ma_2$

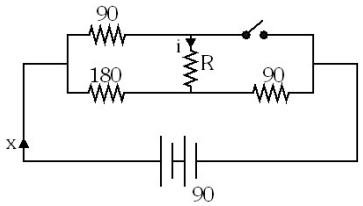
From (3) & (5) $\frac{3F}{5} - \frac{10}{4}ma_p = ma_1$

using (1) in above relation we get $a_P = \frac{2F}{11m}$

22.



Let say current by battery is x and current in bulb is i



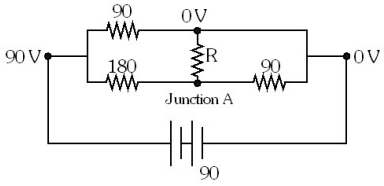
$$x = \frac{90}{\left(\frac{(90+R)\times 180}{270+R}\right) + 90} = \frac{270+R}{450+3R} \qquad \ldots(1)$$

Since (90 + R) and 180 are in parallel

$$i(90+R) = (x-i)180 \qquad i = \frac{180x}{270+R} \qquad \ldots(2)$$

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Solving eq. (1) & (2) $i = \dfrac{60}{150+R}$



Let say potential of junction A is B and apply Kirchhof's junction law for A

$$\frac{90-V}{180} = \frac{V-0}{R} + \frac{V-0}{90} \qquad V = \frac{\frac{1}{2}}{\frac{1}{180}+\frac{1}{90}+\frac{1}{R}}$$

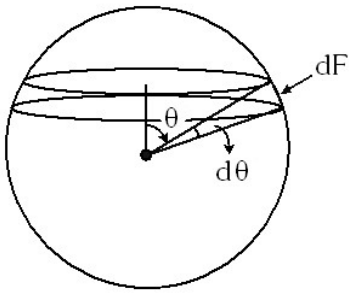
Current through bulb in first condition i = $\dfrac{60}{150+R}$

and current through bulb in second condition i = $\dfrac{V}{R}$

$$\frac{60}{150+R} = \frac{V}{R} = \frac{\left(\frac{\frac{1}{2}}{\frac{1}{180}+\frac{1}{90}+\frac{1}{R}}\right)}{R}.$$ Solving we will get R = 30

23. The number of molecules passing through a hole of cross-sectional area A during a time interval t is $AvNt$, where v is the component, perpendicular to the wall, of the average molecular velocity, and N is the number density (the number of particles in unit volume). At equilibrium no of molecules going out should be equal to the number of molecules going in. The square of the speed of the molecules (a measure of the internal energy of the gas) is proportional to the gas temperature T, and, from the ideal gas equation, the number density proportional to the quotient of the pressure p and the temperature.

24.



$$dF = (Rd\theta)(2\pi R\sin\theta)\left(\frac{R}{2} - R\cos\theta\right)\times\rho g$$

$$F_y = \int_{\pi/3}^{\pi} dF\cos = \frac{9}{8}\pi R^3\rho g$$ (Total upward buoyancy force)

At equilibrium buoyancy force = weight of the floating object

25. $\dfrac{\Pr^3}{R^3} = kA\dfrac{dT}{dr} = k.4\pi r^2\dfrac{dT}{dr}$ $\dfrac{P}{R^3}\int rdr = 4\pi r\int dT$ $\dfrac{PR^2}{2R^3} = 4\pi k\left(T_C - T_S\right)$ $\Rightarrow \dfrac{P}{2R} = 4\pi k\Delta T$

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$$\Rightarrow P = 8\pi kR\Delta T = eA\sigma.T_S^5 \Rightarrow \Delta T = \frac{4\pi R^2\sigma.T_S^4}{8\pi kR} \Rightarrow \Delta T = \frac{\sigma.T_S^4}{2kR} = \frac{\sigma RT_S^4}{2k} \Rightarrow m=1, n=4, \ell=2$$

26. $\beta = \frac{\lambda P}{d} = \frac{5000\times10^{-10}\times10}{\sqrt{2}\times10^{-3}} = \frac{5}{\sqrt{2}}\times10^{-3}m$

Now, since the screen can be assumed to be

Inclined at 45^0 w.r.t slits, hence fringe width on the screen

$$\frac{\beta}{\cos\theta} = \frac{5}{\sqrt{2}}\times10^{-3}\times\frac{1}{\cos45^0} = 5\times10^{-3}m$$

27. F_{net} due to radiation $\frac{I\pi R^2}{C}$ in the direction of radiation. $\therefore F_H = \frac{I\pi R^2}{C}\cos60^0 = \frac{I\pi R^2}{C}$

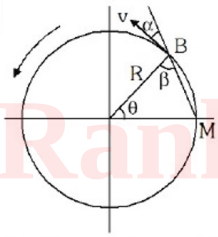
This force is balanced by spring force. $\therefore Kx = \frac{I\pi R^2}{2C} \Rightarrow x = \frac{I\pi R^2}{2KC} = 5$

28. Natural length = 3.45 cm

K(0.04) = 10 K = 250 N/cm Now $K' = \frac{250}{1.15}(3.45) = 750$

$$750\frac{(5-x)}{100} = 30 \quad 5-x = \frac{3000}{750} = 4 \quad x = 1$$

29. Let the position of Buzzer at time t be B and that of the mic be M.



$$V = \omega R; \quad \theta = \omega r; \quad \beta = \frac{\pi-\theta}{2} \quad \alpha = \frac{\pi}{2}-\beta = \frac{\theta}{2} = \frac{\omega t}{2}$$

$$\therefore f = f_0\left[\frac{V_0}{V_0+V\cos\alpha}\right] = f_0\left[\frac{V_0}{V_0+\omega R\cos\left(\frac{\omega t}{2}\right)}\right]$$

30. Change in momentum of I particle = $2Mv_0\cos\theta$

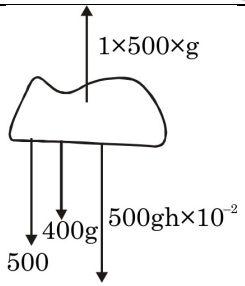
No. of particle striking per second per unit length $= (nv)Rd\theta\cos\theta$

$$f = \int_{-\pi/2}^{\pi/2}(nvR\cos\theta)d\theta[2mv\cos\theta]\cos\theta = 4nmv_R^2\int_0^{\pi/2}\cos^3\theta d\theta$$

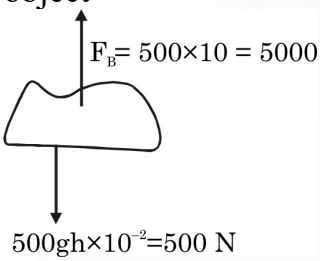
$$= \frac{nmv^2R}{4}\int_0^{\pi/2}(\cos3\theta+3\cos\theta)d\theta = nmv^2R\left[\frac{\sin3\theta}{3}+3\sin\theta\right]_0^{\pi/2} = \frac{8nmv^2R}{3}$$

31.

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$500g = 500 + 500gh \times 10^{-2} + 4000$, $500g = 50h \Rightarrow h = 10m$, Force by liquid on upper object



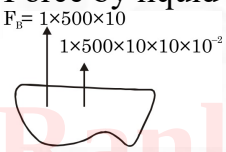
$5000 - 500 = 4500N$

Force by liquid on lower object

$500gh \times 10^{-2} = 500\ N$

$5000 - 50 = 4500\ N$

Force by liquid on lower object

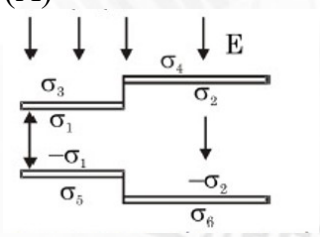


$5000 + 500 = 5500$

Total force by liquid on both object

$5500 + 4500 = 10^4\ N$

32. (A)



$\sigma_1 A_1 + \sigma_2 A_2 = -(\sigma_3 A_1 + \sigma_4 A_2)$ $\sigma_1 A_1 + \sigma_2 A_2 = (\sigma_5 A_1 + \sigma_6 A_2)$

$2(\sigma_1 A_1 + \sigma_2 A_2) = (\sigma_5 - \sigma_3)A_1 + (\sigma_6 - \sigma_4)A_2$

$\frac{\sigma_1}{\epsilon_0} = E_1, \frac{\sigma_2}{\epsilon_0} = E_2, \frac{(\sigma_5 - \sigma_3)}{2\epsilon_0} = E, \frac{(\sigma_6 - \sigma_4)}{2\epsilon_0} = E$

$2A_1E_1\epsilon_0 + 2A_2E_2\epsilon_0 = 2\epsilon_0 EA_1 + 2\epsilon_0 EA_2$

$A_1E_1 + A_2E_2 = E(A_1 + A_2)$ ..... (1)

$E_1d_1 = E_2d_2$ ..... (2)

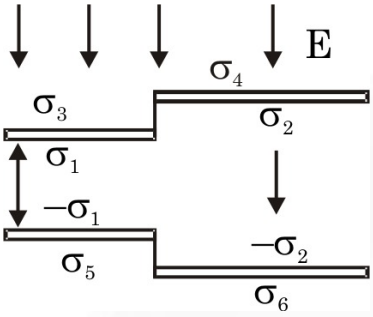
$E_1 = Ed_2 \frac{(A_1 + A_2)}{A_1d_2 + A_2d_1} = \frac{100 \times 2(1+2)}{(2 + 2 \times 1)} = \frac{200 \times 3}{4} = 150$

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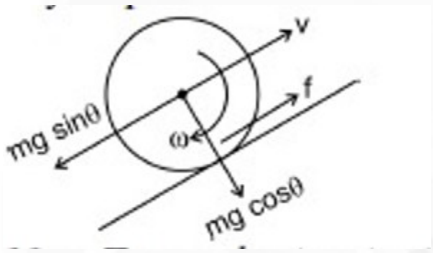
$$E_2 = Ed_1\frac{(A_1+A_2)}{(A_1d_2+A_2d_1)} = 75$$

$$E_1 + E_2 = 225$$

$$E_1 - E_2 = 75$$



33. (A) For pure rolling motion $v = \omega R, mg\sin\theta$ will reduce v so ω must reduce, $mg\sin\theta$ & $mg\cos\theta$ will not produce any torque. So friction should be up the incline.

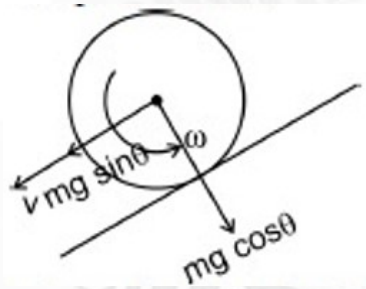


Now, torque about centre $\neq 0$

Torque about point of conctact $\neq 0$

$A \rightarrow P$

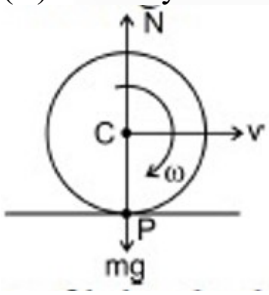
(B) Again for pure rolling $v = \omega R$ should be valid, v will $\uparrow$ with time so ω shold be $\uparrow$. So friction should be up the incline.



Torque about centre $\neq 0$

Torque4 about point of contact $\neq 0$

(C) Velocity v will not change with time so ω should not change.



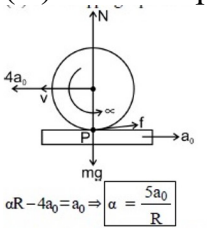
So friction shold not be present $\tau_C = 0$

$\tau_C = 0 \Rightarrow$ Angular momentum is conserved about C.

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$\tau_P = 0 \Rightarrow$ Angular momentum is conserved about P. $C \rightarrow S,R,T$

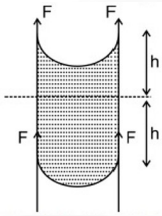
(D) For no slipping at pint ofcontact



$\alpha R - 4a_0 = a_0 \Rightarrow \boxed{\alpha = \frac{5a_0}{R}}$

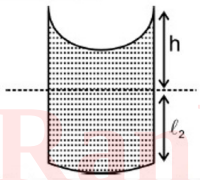
For this angular acc. Torque shold be present about centre. So friction should be in opposite direction of velocity. Torque about P is zero $\Rightarrow$ Angular momentum is conserved about point P. $D \rightarrow Q,R$

34. (A) As liquid rises upto light h due to surface tension which means force of surface tension can lift or handle liquid upto light h



So, Hight of liquid is 2h & Both surface are hemispherical. $A \rightarrow P,Q$

(B) As $\ell_1 > h \Rightarrow$ Length of liquid above liquid surface is h



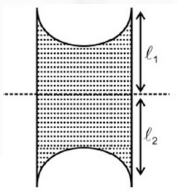
But $\ell_2 < h \Rightarrow$ Length of liquid in capillary below liquid surface is ℓ_2

Overall length in liquid in capillary $= \ell_2 + h$

Upper surface $\rightarrow$ Hemispherical

Lower surface $\rightarrow$ A little flat with respect to hemispherical surface $B \rightarrow T$

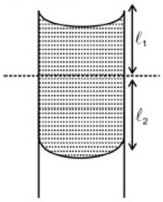
(C) In absence of gravity liquid will fill entire length of capillary $g_{eff} = 0 \Rightarrow h \rightarrow \infty$



As atmospheric pressure is present therefore when we observe liquid surface from top it should appear concave. $C \rightarrow P,R$

(D) $\ell_1 < h \Rightarrow$ Capillary is of insufficient length so liquid will raise upto length ℓ_1 but surface is not fully hemispherical.

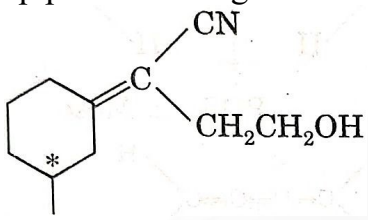
As $\ell_2 < h \Rightarrow$ Liquid will be in all length but surface is not fully hemispherical.



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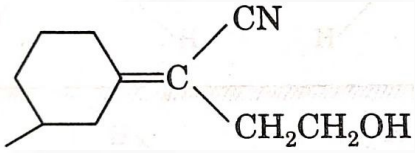
CHEMISTRY

35. (a) Two different geometrical isomers are possible due to relative position of $-CH_3$ group present in ring.



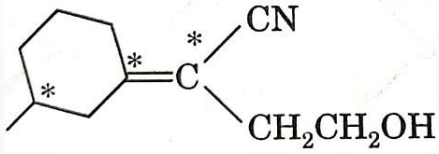
(b)

Optical active one chiral carbon is present



(c)

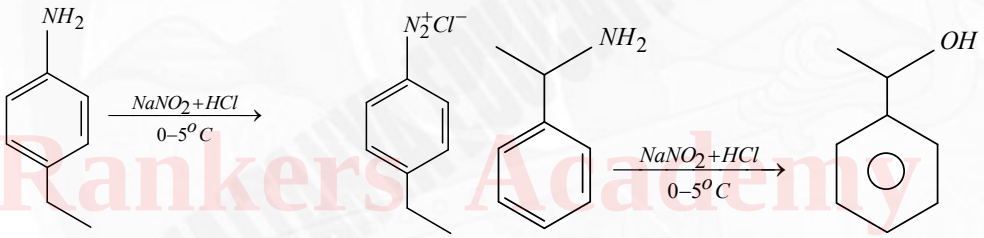
Senior groups are on opposite side

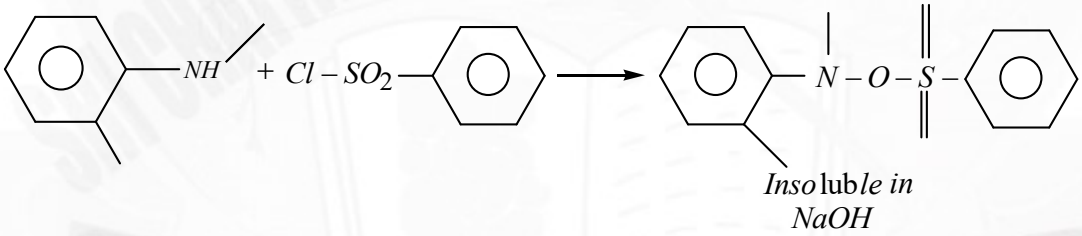


(d)

Three stereocentre

36.



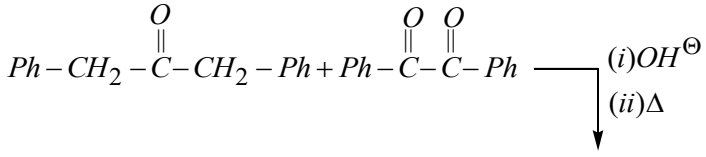


37. $LiAlH_4$ & dibal – h, B_2H_6, can reduces carboxylic acid as well.

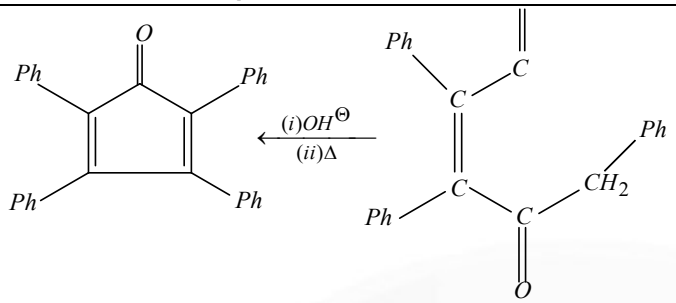
38. $P = BF_3, Q = BCl_3, R = B_2H_6, S = B(OH)_3$

39. Among the observations:
(i) Since chromyl test is coming positive, it suggests anion can be chloride ion.
(ii) Brown ring test is positive, but intense brown fumes not appearing on addition of Cu turnings suggests the anion can be nitrite.
(iii) Violet layer in layer test suggests anion can be iodide.

40.



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41. A more reactive metal can displace a less reactive metal from its salt solution.
Metal Q: Displaces all four metals (Ag, Pb, Ni, and Zn). This means Q is the most reactive of the group.

42 $K_{sp}\left[Zn(OH)_2\right]=\left[Zn^{+2}\right]\left[OH^-\right]^2 \qquad 2\times10^{-13}=0.2\times\left[OH^-\right]^2$

$\left[OH^-\right]=10^{-6} \qquad K_{sp}\left[Al(OH)_3\right]=\left[Al^{+3}\right]\left[OH^-\right]^3$

$16\times10^{-25}=\left[Al^{+3}\right]\left[10^{-6}\right]^3 \qquad \left[Al^{+3}\right]=16\times10^{-7}$

43. $X_B=0.5\,X_T=0.5\;\; P_{total}=P_B^0X_B+P_T^0.X_T$

$80=0.5P_B^0+0.5P_T^0 \qquad 160=P_B^0+P_T^0\;\ldots\ldots\ldots(1)$

$y_B=\dfrac{P_B^0.X_B}{P_{total}}=\dfrac{P_B^0\times0.5}{80}=\dfrac{P_B^0}{160} \qquad y_T=\dfrac{P_T^0.XT}{P_{total}}=\dfrac{P_T^0\times0.5}{80}=\dfrac{P_T^0}{160}$

For condensate

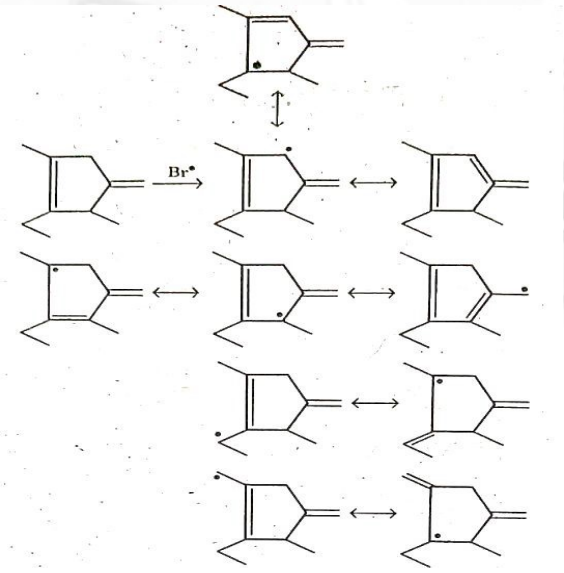
$P_{total}=100=P_B^0\cdot y_B+P_T^0.y_T \qquad =100=p_B^0\cdot\dfrac{P_B^0}{160}+P_T^0.\dfrac{P_T^0}{160}$

$P_B^{0^2}+P_T^{02}=100\times160\;\ldots\ldots..(2)$

Form (1) and (2)

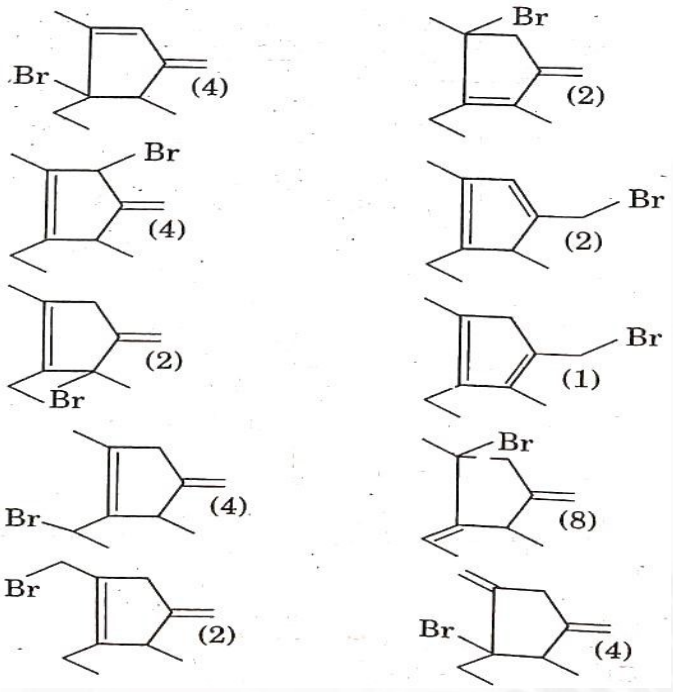
$P_B^0=120 \qquad P_T^0=40$

44. 10 intermediate free radicals.



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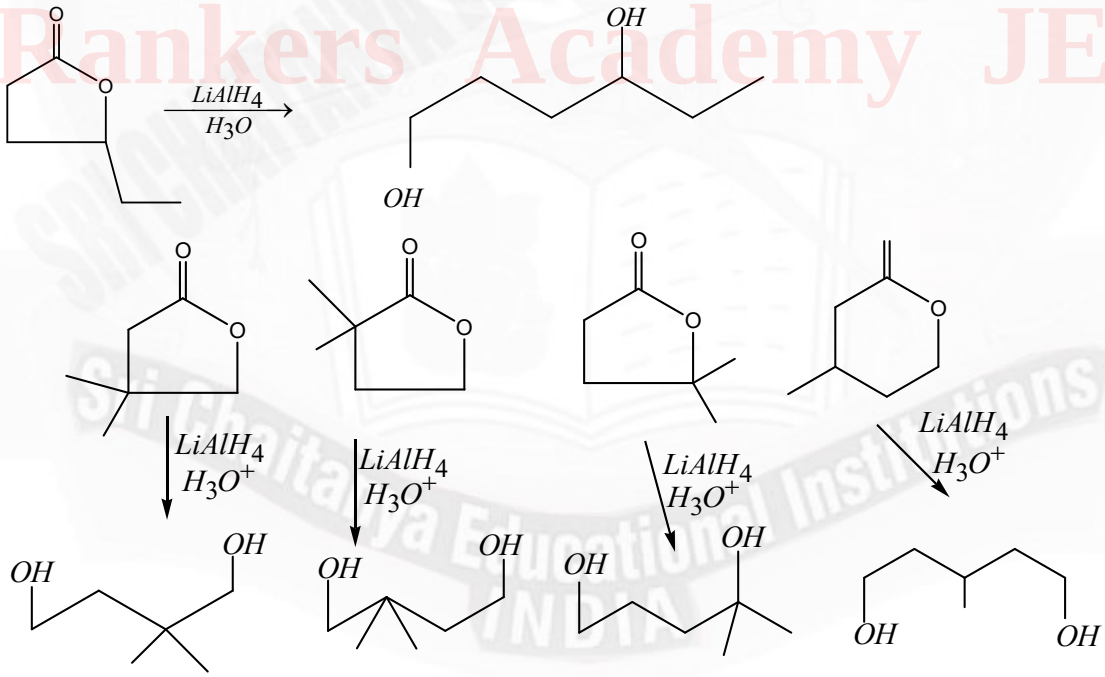
So 10 structural isomeric products are formed and total products including stereoisomers are 33.



45. $E^{o}_{cell} = 151 - 0.78 = 0.73V$

$$X = 0.73 + \frac{0.06}{5}\log\left[H^{+}\right]^{8} = 0.73 - 0.48 = 0.25$$

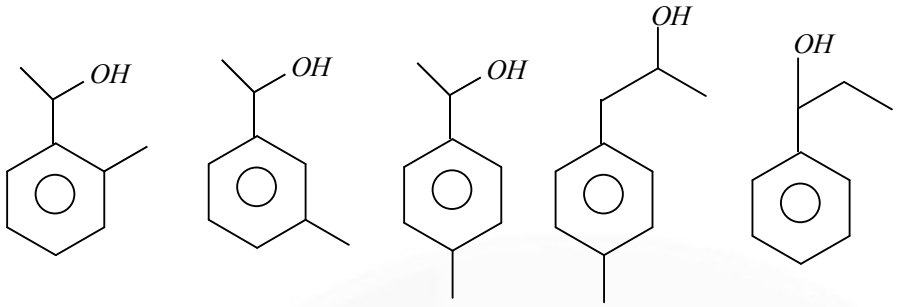
46.



Above dihydroxy alcohols are achiral

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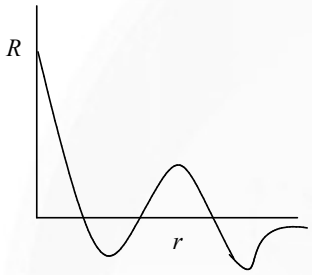
For More Join: @IITJEE_Advanced



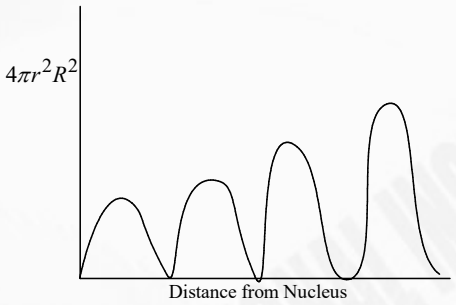
47.

And their enantiomers

48. $A \to P; B \to PQS; C \to QS; D \to QS$



-4s,



-4s, $5p_y$, $6d_{xy}$

Angular probability depends upon θ and ϕ - $5p_y$, $6d_{xy}$

Atleast one angular node is present- $5p_y$, $6d_{xy}$

49. A) Chloroform + Aniline $\to$ (III) Distillation
B) Benzoic + Napthalene $\to$ (IV) Crystallisation
C) Water + Aniline $\to$ (I) Steam distillation
D) Napthalene + Sodium chloride $\to$ (II) Sublimation

50. A) $CH_3 - CHBr - CD_3$ on treatment with alc. KOH gives $CH_2 = CH - CD_3$ as a major product - E2 reaction
B) $Ph - CHBr - CH_3$ reacts faster than $Ph - CHBr - CD_3$ - E2 reaction
C) $Ph - CH_2 - CH_2Br$ on treatment with $C_2H_5OD / C_2H_5O^-$ gives $Ph - CD = CH_2$ as the major product - E1CB reaction
D) $PhCH_2CH_2Br$ and $PhCD_2CH_2Br$ react with same rate - E1 reaction, First order reaction

51. $I \to RT; II \to PSRT; III \to QT; IV \to PQ$

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